

BLOCKCHAIN BASED CROWDFUNDING FOR EDUCATION

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Abstract:

This research paper presents a pioneering approach to revolutionise education funding through a blockchain-based crowdfunding platform, designed to empower students, educators, and educational organisations. Our objective is to provide a decentralised space where innovative educational initiatives can be funded, enabling financially challenged students to seek support, and facilitating small schools' appeals for infrastructure and program enhancements. In response to the growing demand for transparency and accountability in crowdfunding, our platform fosters collaboration among developers to enhance project features and capabilities while implementing a user-friendly web interface using Web3 technology. The primary focus is on promoting community engagement and garnering support for educational projects while evolving and expanding the platform's capabilities with the scalability advantage of polygon.

Keywords: Blockchain, Decentralised , Crowdfunding , Polygon, Web3, Smart Contracts.

1. INTRODUCTION

In the contemporary landscape, a multitude of students worldwide face hurdles on their path to higher education due to limited financial resources. The dream of pursuing studies is shattered as the cost of education rises, preventing many young individuals from affording tuition, textbooks, and living expenses essential for attending college or university. This financial barrier not only stifles ambitions but also narrows opportunities for personal and professional growth.

To address this challenge, a crowdfunding platform has been developed using blockchain technology specifically for students. This innovative approach leverages blockchain and smart contracts to tackle financial constraints. Through our platform, students can engage in crowdfunding initiatives, securing funds for education and realising dreams of higher learning.

The proposed blockchain technology operates as a decentralised, distributed digital ledger recording transactions across an internet-based computer network. This decentralisation makes it challenging to alter or manipulate recorded information. The crowdfunding technique involves generating funds by collecting small contributions from a substantial number of individuals through online platforms and social media, providing a powerful tool for supporting educational endeavours.

In contrast to conventional crowdfunding methods relying on intermediaries like banks, financial institutions, payment gateways, and crowdfunding platforms, our approach minimises dependencies. By eliminating intermediaries, we aim to reduce fundraising costs and empower project managers with greater control over their initiatives. This streamlined process enhances the efficiency and effectiveness of crowdfunding for education, making it a more accessible and impactful avenue for aspiring students.

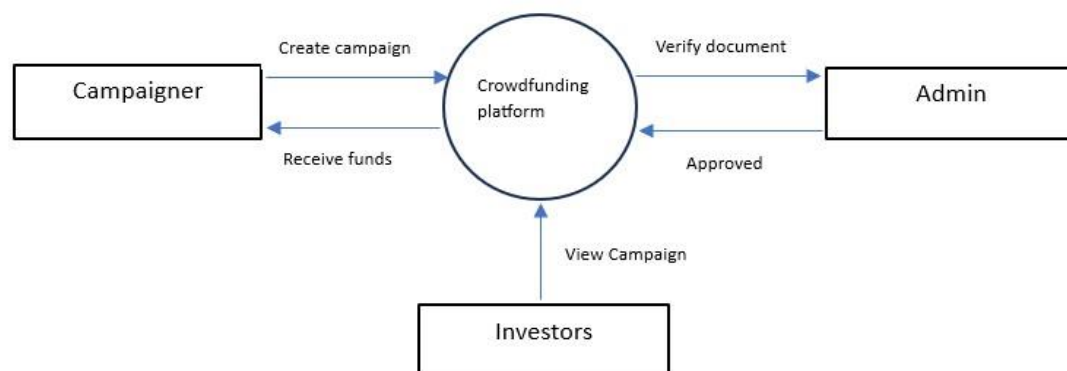


Figure 1. Process of crowdfunding

With the help of blockchain technology, this proposed method can solve these issues by taking advantage of the special qualities of the blockchain, like security, decentralisation, and transparency. Blockchain technology offers an immutable, decentralised ledger that makes transaction tracking safe and transparent, without requiring any middlemen.

Some studies related to blockchain-based crowdfunding show how crowdfunding helps startups, businesses, and individuals to raise funds without much problem. However, these studies mainly focus on the crowdfunding process and do not apply it for education purposes. In a previous research paper [9] in which fundraisers are used as intermediaries, which is the main disadvantage of that particular paper. This proposed method resolves this problem and enhances the performance of the platform, making it effective for both donors and fundraisers.

2. LITERATURE REVIEW

The author's analysis delves into current challenges within China's equity crowdfunding industry. This paper explores the practical applications of integrating blockchain technology into equity crowdfunding, drawing insights from a thorough examination of its attributes.

The use of blockchain technology presents a potentially secure, efficient, and cost-effective method for recording stocks and shares of companies financed through crowdsourcing.

Facilitating easier transfers and transactions of crowdsourced equity, blockchain technology enhances the circulation of stocks. It empowers peer-to-peer transactions between investors and entrepreneurs, addressing concerns related to fund management security and regulatory compliance.

Furthermore, blockchain technology can be harnessed to establish a voting system for crowd funders, providing them with an avenue to actively participate in corporate governance. This multi-faceted approach not only addresses current issues within China's equity crowdfunding industry but also lays the foundation for a more streamlined, secure, and participatory crowdfunding ecosystem. This aids in defending the rights and concerns of small investors [1].

A crowdfunding platform built on blockchain technology with a token system for investment was presented by Yurtsever and Karaçalı. They contend that the use of tokens can lower the risk of fraud and boost investment liquidity. Through the use of their technology, investors may purchase tokens using cryptocurrency. These tokens can subsequently be exchanged on cryptocurrency platforms, giving investors access to a secondary market. Their platform is not user-centric. This is the main drawback of earlier work [2].

By reviewing previously published research on crowdfunding and blockchain technology, the author of this study presents a conceptual framework that may be able to address issues with crowdfunding contracts by utilising blockchain technology. They investigate how the fundamental features of blockchain technology could help these platforms overcome their issues while accounting for the involvement of the intermediate platforms. They talk about the ways that blockchain technology can manage the connection between platforms, donors, and fundraisers in a safe and efficient manner. They hope that by publishing their work, scholars will be more interested in exploring the real-world uses of utilising blockchain technology in crowdfunding agreements [3].

Laporte et al. propose the utilisation of blockchain technology for crowdfunding purposes with an emphasis on guaranteeing transparency and lowering intermediation costs. They contend that third parties like banks, payment processors, and crowdfunding platforms have the power to raise the price of crowdsourcing and limit project founders' ability to manage their fundraising efforts. Crowdfunding costs can be decreased and intermediaries removed with the use of blockchain technology [4].

The authors (Kshetri et al.) suggest a crowdsourcing platform built on blockchain that distributes money via smart contracts. Funds are automatically disbursed to project creators via the smart contract subject to predetermined criteria, such as meeting funding targets or project milestones. This reduces crowdsourcing expenses and enhances transparency by doing away with the need for middlemen like banks or crowdfunding sites [5].

Medeiros and Santos proposed a blockchain-based decentralised crowdfunding platform. They contend that decentralisation can improve transparency while decreasing the need for middlemen. Their technology, in particular, makes use of a decentralised application developed on top of the Ethereum blockchain. The dApp enables project developers to launch crowdfunding campaigns and backers to donate to such projects with bitcoin. When the financing target is reached, the money is automatically released to the project author via a smart contract [6].

The platform developed by Tsygankov et al. supports the objectives of lower intermediation costs and more transparency. Their method, however, ignores the requirement that project milestones be voted on in order to guarantee accountability. This can be a drawback of their platform as, once financing is obtained, backers would not be able to influence the project's course [7].

Yao et al. propose a blockchain-based smart contract-based crowdfunding platform. They contend that smart contracts can increase transparency while also automating the crowdfunding process. Their platform, in particular, enables project creators to launch a financing campaign by writing a smart contract that specifies project specifics, funding objectives, and fund distribution. Backers can donate to the campaign once the contract is launched on the blockchain by transferring bitcoin to the contract's address. When the financing target is fulfilled, the smart contract immediately delivers the cash to the project creator, or returns the funds to the supporters in case the objective is not accomplished within a designated time frame. Although Integrating smart contracts into crowdfunding can reduce the reliance on intermediaries and enhance transparency, Yao et al. 's method does not take into consideration the requirement for project milestone voting to guarantee accountability [8].

The author described in this paper a system utilising blockchain technology for generating and storing agreements among students and their higher education sponsors, facilitated by intermediary agents commonly referred to as fundraisers. The sponsorship could take the form of a scholarship, contribution, or loan. A group of competitive fundraisers will maintain, organise and administer the money will hold the distributed ledgers and operate as the miners in the blockchain network. Underlying our job is a just cause: helping underprivileged kids. Gifted children shouldn't be denied the opportunity to pursue higher education because their family is unable to provide financial assistance. Through the outsourcing of funding for higher education that these students could not afford on their own, this platform will open doors for the less fortunate pupils [9].

In this paper the author describes security and cryptocurrency. Project owners maintain and initialise the cryptocurrency which acts as the share in the project of the owner. By which transparency in the fundraiser and the donors are maintained. The currency is bought by the donors in the owner's project and helping the project owners or fundraisers. Indirectly the donors can help the project creator by buying the cryptocurrency shares [10].

A reputation-based mechanism is used by Li and Li to manage crowdfunding projects in their blockchain-based crowdfunding system. Their system seeks to address the informational problem of crowdfunding asymmetry through the use of reputation scores aimed for project initiators. They contend that by using scores, supporters are better equipped to decide what supports initiatives aimed at lowering the risk of fraud. Their method doesn't deal with the issues of accountability and transparency in crowdfunding. Backers have no control over how their money is spent, and there is no way for the system to guarantee that project creators are held responsible for how they spend the money. Furthermore, if project developers are able to artificially boost their reputation rankings, the system may be subject to manipulation [11].

After reviewing the previous proposed method and techniques, it is noticed that some issues are found regarding security and authenticity. So here in this method or proposed module try to resolve all the issues and make the platform effective and user centred. Implement voting mechanism, smart contract, blockchain technology and eliminate the intermediary person which is good to campaigner and investors.

3. METHODOLOGY

According to previous research papers and survey papers, blockchain technology has been introduced to crowdfunding platforms. However, these methods have some drawbacks. This paper identified and resolved these drawbacks in the proposed method, which introduces how to resolve these problems and achieve the goal from the user's perspective. The platform includes the following methodology.

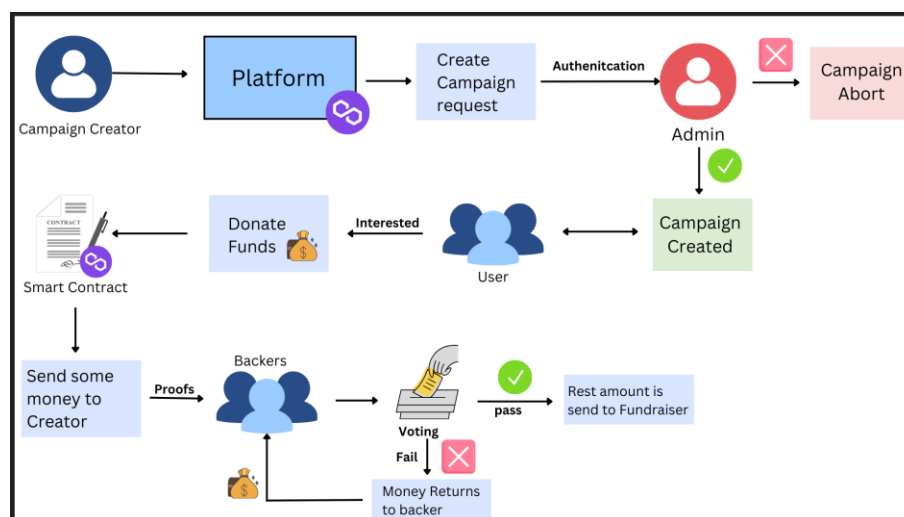


Figure 2. Proposed Methodology

Firstly, campaigners (students, users, etc.) reach out to the platform and read all of the rules to create a campaign. Following that, individuals proceed to create their own campaign and upload pertinent project details, qualifications, income certificates, and results.

Subsequently, an administrator meticulously reviews the submitted documents and certificates. If the administrator verifies all the details and concurs with them, approval is granted for the project. Conversely, if any discrepancies arise during the verification process, leading to disagreement with the provided information, the campaign is terminated, and the campaigner is promptly notified of the cancellation. Upon the administrator's approval, the campaign is officially initiated, enabling the campaigner to commence fundraising for their project. In the subsequent phase, users have the opportunity to visit the platform and assess the campaign. If satisfied with the project details and intrigued by the campaigner's initiative, users can contribute funds to the project. It is imperative to note that all funds are securely held in smart contracts, with withdrawal restrictions in place. Only the creator possesses the authority to withdraw funds post the voting phase. The voting process, a pivotal component of this method, ensures the secure withdrawal of funds, requiring at least 51% of favourable votes for the campaigner to access the allocated funds.

The effectiveness of the proposed voting mechanism significantly enhances the overall reliability of the process. If the vote is less than 51%, then it means that the campaigner or creator cannot withdraw the amount. After that, the funds donated by the users will be returned. This means that users are not hesitant to donate funds to the project. The whole system is fully secure and authenticated. Here, campaigner means all students and educational institutions who can create campaigns to raise funds for their education, infrastructure, projects, etc. So overall, the proposed platform is effective and user-friendly, and it is easy to understand. All of the steps of the methodology are included in the figure above.

4. COMPARATIVE STUDY

Comparing the results of previous research papers with this paper. That is based on the technology, approaches, user's view and many more.

Table 1. Comparing the proposed method to prior methods.

Author's Name	Methods of Previous Authors	Year	Proposed Method
Zhu and Zhou [1]	Equity based crowdfunding platform.	2016	Donation based crowdfunding platform.
Yurtsever and Karaçalı [2]	Provide a token mechanism.	2018	Provide voting mechanisms.
H. Zhao and C.P.K. Coffie[3]	Implementation of blockchain technology	2018	Implementation of blockchain in addition to smart contracts.
Laporte et al. [4]	Don't focus on users authenticity	2019	Focus on user authenticity by verifying documents.
Kshetri et al. [5]	Emphasise how smart contracts may be used to allocate funds and increase transparency.	2019	Emphasise the work of smart contract in more efficient way
Medeiros and Santos [6]	Use Ethereum blockchain	2019	The requirement of a voting mechanism is introduced.
Yao et al. [8]	Did not address the need of voting and smart contract mechanism	2020	Address the need of voting and smart contract mechanism
Mahmood A. Rashid, Krishneel Deo and 4 more.[9]	Intermediary persons are included	2020	Intermediary persons are not included
Li and Li [11]	Proposed method does not meet the difficulties of crowdsourcing transparency and accountability.	2021	Proposed method meets all needs authenticity and transparency

CONCLUSION

Crowdfunding is one of the technologies that has made it easy for new platforms to help startups and organisations raise funds through traditional means. Our main goal is to support talented students who are willing to pursue further education but face financial constraints. The platform offers an opportunity to underprivileged students by raising funds for their studies. On this platform, fundraisers will present their project ideas along with all the related documents, which should be accessible to all team members. Funders who support the project idea are essentially purchasing shares in the project and providing their support. With the implementation of blockchain technology within crowdfunding, the amount of paperwork will be reduced, and all transactions will be recorded, enhancing trust and reliability in the project idea. Furthermore, documents will be secure and easily accessible. One of the most significant advantages of using blockchain technology is that every funder can track the daily progress of a project. In the future, more sophisticated smart contracts can be developed to handle a wider range of crowdfunding scenarios, such as equity crowdfunding, debt crowdfunding, and reward-based crowdfunding.

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